Defence Guide · Student 2

Student 2: Results Health Anxiety, AI Chatbot Use and Cyberchondria Among Adults in Baghdad

> **Your role.** You carry the heaviest numbers. Your job is to make them simple: one message per slide, then the “In summary” line.

You present	Slides 13–23 (≈ 6 minutes)
You own in the thesis	Chapter 3 Results (3.1–3.10), Tables 1–13, Figures 3–11, Appendix A; and Section 2.6 (you are the team's statistics person)
Your handover line	“Those are our results. My colleague will now discuss what they mean.”
Your teammates	Student 1: Introduction & Methods (slides 1–12) · Student 3: Discussion & Conclusions (slides 24–33)

What is in this guide

1. The whole study: two-minute script, core facts, terms, “never say” list (same in all three guides)
2. Your chapter, section by section: what it says, the numbers, the sources, what to watch
3. Your key sources: each reference number in your chapter and what it found
4. Your slides, one by one: what is on screen, your script, timing and traps
5. Questions: on your part, on the whole study, and difficult ones, with model answers
6. Editing your chapter safely and in your own words
7. Your 14-day plan
8. Self-test with answers
9. Checklist for the day before

Part 1 · The whole study (everyone must know this)

> **The committee can ask any of you about any part.** You split the presentation, not the knowledge. Learn this part so well that you can explain the whole study without slides.

The study in two minutes (practise saying this out loud)

"Many people now look up their symptoms online before seeing a doctor, and more and more of them ask AI chatbots like ChatGPT. For some people this searching makes them more worried, not less. That is called cyberchondria, and it is closely linked to health anxiety. No one had studied this in Baghdad, and AI chatbots had hardly been studied next to search engines. We ran an online survey of 300 adults in Baghdad between 17 July 2026 and 11 September 2026, using two established scales: the SHAI-14 for health anxiety and the CSS-12 for cyberchondria. We found that AI chatbots are now the main source of health information: 78.0% ticked them, more than search engines (56.3%). About one in five (19%) had elevated health anxiety. Health anxiety and cyberchondria were moderately linked, r = 0.47, and this held after adjusting for everything else. The link ran mainly through excessive searching and post-search distress, which fits a vicious cycle, although one survey cannot prove direction. We found no reliable evidence that AI itself adds to cyberchondria beyond health anxiety. And searching shapes the consultation: about a third delay care because they feel reassured, and about a third never tell their doctor what they searched. So our main message for clinicians is simple: ask patients, without judgement, what they searched or asked an AI."

Core facts (memorise; examiners test these)

Topic	Fact
Design	Analytical cross-sectional online survey; STROBE and CHERRIES checklists
Who and when	**300** adults, Baghdad, 17 July 2026 – 11 September 2026; convenience + snowball sampling
Sample size	**194** needed for r = 0.20 (α 0.05, 80% power); margin of error ±5.7%
Sample profile	62.3% aged 20–29 · 56.3% female · **44.0%** medical students · 63.3% health background · 21.7% chronic condition · 70.7% answered in Arabic
Tools used	AI chatbots **78.0%** ticked (88.7% incl. named) · search engines 56.3% · medical platforms 18.3% · social media 10.7% · ChatGPT 59.3% · AI without any search engine 41.0%
Motives	convenience 80.0% · "do I need a doctor?" 57.7% · instant reassurance 51.0%
Health anxiety	SHAI mean 12.3 ± 5.8 (0–42) · **19%** elevated (≥ 18; 95% CI 15.0–23.8%) · 0.7% at ≥ 29
Cyberchondria	CSS mean 28.9 ± 8.4 (12–60) · excessiveness highest, compulsion lowest
Reliability	α 0.81 (SHAI), 0.86 (CSS); ω 0.82, 0.87 · CSS-12 4-factor CFA: CFI 0.95, RMSEA 0.070, SRMR 0.053
Arabic vs English	CSS-12 fully (scalar) invariant; SHAI-14 metric invariant; no item DIF after FDR
Main result	**r = 0.47** (95% CI 0.38–0.55); explains 22% of the variance · partial r = 0.48
By facet	excessiveness 0.46 · distress 0.45 · compulsion 0.25 · reassurance 0.24
Regression	R^2 0.030 (demographics) → 0.133 (+ digital use) → 0.317 (+ health anxiety, ΔR^2 18.4%); β = 0.48
Who is anxious	AOR **3.12** per SD of cyberchondria · 3.83 reassurance motive · 0.61 per older age group · AUC 0.83 → 0.79
Pathways	via excessive searching 0.16 (CI 0.11–0.23) · via distress 0.22 (direct → 0.01) · reverse 0.15
Behaviour	delay care when reassured **34.7%** · ignore symptom 30.0% · several doctors 25.3% · urgent care 22.0% · self-medication 21.3%
Doctor relationship	never tell the doctor **31.7%** · always tell 27.0% · trust strained 34.7% · higher cyberchondria → more disclosure (OR 0.57, q 0.012)

AI effect	+5.3 CSS points (inclusive) but +1.5, p = 0.31 (strict) → no reliable independent effect · 32 named-not-ticked · 34 non-users
Robustness	every sensitivity analysis left the main result unchanged (Table 13)

Terms you must be able to explain in one sentence

Term	Plain explanation
Health anxiety	Persistent worry about having or developing a serious illness, kept going by misreading normal body sensations and by reassurance seeking.
Illness anxiety disorder	The DSM-5 diagnosis at the severe end of health anxiety. We did **not** diagnose it; we measured symptoms.
Cyberchondria	Excessive or repeated online health searching that increases rather than relieves distress.
SHAI-14	14 items, each scored 0–3 (total 0–42); higher = more health anxiety. ≥ 18 = elevated (screening threshold).
CSS-12	12 items scored 1–5 (total 12–60) with four 3-item facets: excessiveness, compulsion, distress, reassurance seeking.
Cross-sectional	Everything measured once, at one time point: shows associations, not cause and effect.
Convenience / snowball sampling	People who were easy to reach, who then shared the link with others. Quick, but not representative.
Mean ± SD	The average score and how spread out the scores are.
Median (IQR)	The middle score and the range of the middle half; used when data are skewed.
p-value	How surprising the result would be if there were truly no effect; $p < 0.05$ is called significant.
95% CI	The range in which the true value probably lies; narrower = more precise.
Correlation r	How closely two scores move together (0 = none, 1 = perfect). 0.47 is moderate.
Partial correlation	The same link after removing the effect of other variables.
Mann–Whitney / Kruskal–Wallis	Non-parametric tests comparing two / three or more groups, used because scores were not normally distributed.
χ^2 test	Tests whether two categorical variables are related (e.g., social-media use and elevated health anxiety).
FDR (q-value)	Correction for running many tests; a result is "robust" if $q < 0.05$, otherwise exploratory.
Linear regression, B and β	Predicts a score from several factors at once. B = change in CSS points per unit; β = the same on a standardised scale (to compare factors).
R^2 and ΔR^2	Share of variation explained by the model; ΔR^2 = extra share explained by adding a block of variables.
Logistic regression, AOR	Predicts a yes/no outcome. Adjusted odds ratio: how many times the odds change, holding other factors constant.
AUC	How well a model separates two groups (0.5 = chance, 1 = perfect).
Mediation	Splitting a link into a part that runs through a third variable (indirect) and the rest (direct). Statistical, not causal.
Moderation	Whether a third variable changes the strength of a link (e.g., does AI use make the link stronger? No).
Cronbach α / McDonald ω	Internal consistency: do the items of a scale hang together? ≥ 0.7 is acceptable, ≥ 0.8 good.
CFA (CFI, RMSEA, SRMR)	Checks the items group into the intended factors. Good: CFI ≈ 0.95+, RMSEA ≤ 0.08, SRMR ≤ 0.08.
Measurement invariance	Checks the scale measures the same thing in two groups (here Arabic vs English) before comparing them.

Sensitivity analysis	Re-running the main analysis in a different way to see if the result holds.

Never say (overclaims the committee will catch):

- "Health anxiety *causes* cyberchondria" → say "is associated with", "is compatible with a cycle".
- "19% *have* health anxiety disorder" → "19% scored in the elevated range on a screening tool".
- "AI makes people anxious" → "we found no reliable independent effect of AI use".
- "Our results represent Baghdad" → "a young, online, convenience sample; associations were consistent across subgroups".
- "The Arabic version is validated" → "we tested it: equal reliability and formal invariance, but no formal back-translation".

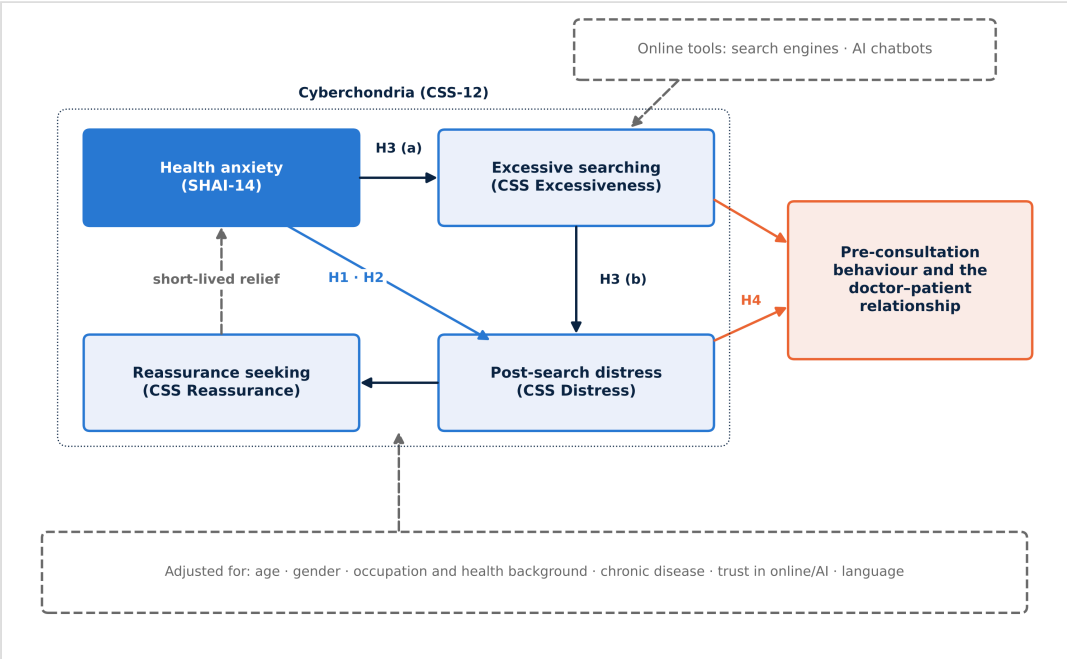


Figure 1 of the thesis: the conceptual framework. Learn to explain every arrow.

Part 2 · Your chapter, section by section

For each section: its **purpose**, the **key points and numbers** (with the reference numbers used in the thesis), and what to **watch**.

3.1 Participants (Table 1)

Who answered.

- 300 adults, no missing data; 62.3% aged 20–29; 56.3% female; most current students; medical students 44.0%; health background 63.3%; chronic condition 21.7% (migraine, hypertension, asthma…); Arabic 70.7%.

Watch: Message: a young, educated, digitally active sample.

3.2 Online health-information seeking (Table 2, Figure 3)

How people search.

- AI ticked 78.0% (88.7% incl. named) vs search engines 56.3%; AI without any search engine 41.0%; ChatGPT 59.3%.
- AI use 96% at 18–19 y, 72% at ≥ 40 y; the 34 non-users are older.
- Motives: convenience 80.0%, “do I need a doctor?” 57.7%, reassurance 51.0%; trust mostly moderate.

Watch: Message: for this sample, the first “doctor” is often an AI.

3.3 Behaviour and the doctor relationship (Figure 4)

- Delay/skip care when reassured 34.7%; ignore 30.0%; several doctors 25.3%; urgent care 22.0%.
- Always tell the doctor 27.0%; never (fear of judgement) 31.7%; relationship complicated or trust reduced 34.7%.

Watch: The “never” option combined frequency and a reason: say “never tell, citing fear of judgement”.

3.4 Levels and psychometrics (Tables 3, 4, 5)

- SHAI 12.3 ± 5.8; elevated 19% (CI 15.0–23.8%); Table 4: 32.7% at ≥ 15, 19.0% at ≥ 18, 7.3% at ≥ 22, 0.7% at ≥ 29.
- CSS 28.9 ± 8.4; excessiveness 9.66 (highest), compulsion 5.00 (lowest).
- Reliability α 0.81 / 0.86; ω 0.82 / 0.87. CSS-12 four-factor CFA (ML) CFI 0.95, RMSEA 0.070, SRMR 0.053; SHAI one-factor poor (CFI 0.83): not strictly unidimensional.
- Table 5: CSS-12 configural → metric ($\Delta\chi^2$ 5.52, p 0.70, ΔCFI +0.002) → scalar ($\Delta\chi^2$ 8.41, p 0.40, ΔCFI −0.001): fully invariant. Both baselines are imperfect (configural CFI 0.937 CSS, 0.808 SHAI), so the SHAI result is hard to interpret. SHAI-14 metric supported (p 0.73), scalar not (p 0.007, ΔCFI −0.018). No item DIF after FDR.

Watch: Explain why prevalence depends on the cut-off and why that is not a weakness: it is honest reporting.

3.5 Main association (Table 6, Figures 5, 6)

The central result.

- r = 0.47 (95% CI 0.38–0.55), ρ = 0.47, explains 22%; partial r = 0.48.
- Facets: excessiveness 0.46, distress 0.45 (stronger, Williams' t p < 0.001) vs compulsion 0.25, reassurance 0.24.
- Item level: fear of serious illness and interpretation of sensations matter, mere awareness does not; 111/168 pairs survive FDR (58 Holm); strongest ρ 0.43 (Figure 11, Appendix A).

Watch: H1 and H2 are answered here.

3.6 Bivariate factors (Tables 7, 8, Figure 7)

162 tests.

- Health anxiety higher with chronic condition (p 0.004, q 0.028) and reassurance motive; age/education/occupation borderline (exploratory); gender no difference.
- AI users higher CSS (inclusive), mainly excessiveness and reassurance; highest in AI + search engine users; chatbot brand no difference.
- χ^2: elevated HA → 52.6% vs 23.0% high cyberchondria (OR ≈ 3.7).

Watch: Remember q-values: “survives correction” vs “exploratory”.

3.7 Multivariable models (Tables 9, 10, Figure 8)

- Hierarchical regression: R^2 0.030 → 0.133 → 0.317; health anxiety adds 18.4% (f^2 medium); B = 0.68, β = 0.48.
- AI: +5.3 (CI 2.3–8.3) inclusive vs +1.5, p 0.31 strict; the 32 reclassified had the highest CSS (31.6 vs 24.0 in non-users).
- Reverse model: CSS strongest predictor of SHAI; chronic disease B +1.7 (p 0.029); medical students no longer differ after adjustment.
- Logistic (57 events): AOR 3.12 cyberchondria per SD, 3.83 reassurance motive, 0.61 age; AUC 0.83 → 0.79; ≈ 5 events per variable → exploratory.

Watch: You will be asked to explain AOR, AUC and why it was corrected.

3.8 Behaviour models (Table 11)

- Cyberchondria (not health anxiety) goes with urgent visits (AOR 1.51), less ignoring (0.65), strained relationship (ordinal OR 1.34).
- Only robust behavioural result: higher CSS → lower odds of never disclosing (AOR 0.57, q 0.012). Health anxiety → concealment AOR 1.47, q 0.098 (exploratory).
- Trust → fewer multiple-doctor visits (AOR 0.59).

Watch: H4 only partially supported.

3.9 Pathways (Table 12, Figures 9, 10)

- HA → excessive searching → distress: indirect 0.16 (CI 0.11–0.23), direct 0.30.
- HA → distress → reassurance seeking: indirect 0.22 (CI 0.16–0.29), direct 0.01 (ns).
- Reverse path also significant (0.15); serial path 0.07. No moderator significant after FDR; age trend p 0.07.

Watch: Always end with: “compatible with a cycle, not proof of direction”.

3.10 Sensitivity (Table 13)

- Excluding annotated (r 0.46) or straight-liners (r 0.47), dropping SHAI item 2 (r 0.48), bootstrap CI for B 0.54–0.83, quantile B 0.55–0.85, no quadratic effect (p 0.25), VIF ≤ 3.9, early vs late responders no difference; residuals non-normal → robust SEs.
- Only the AI effect depended on a decision (definition).

How to read your figures

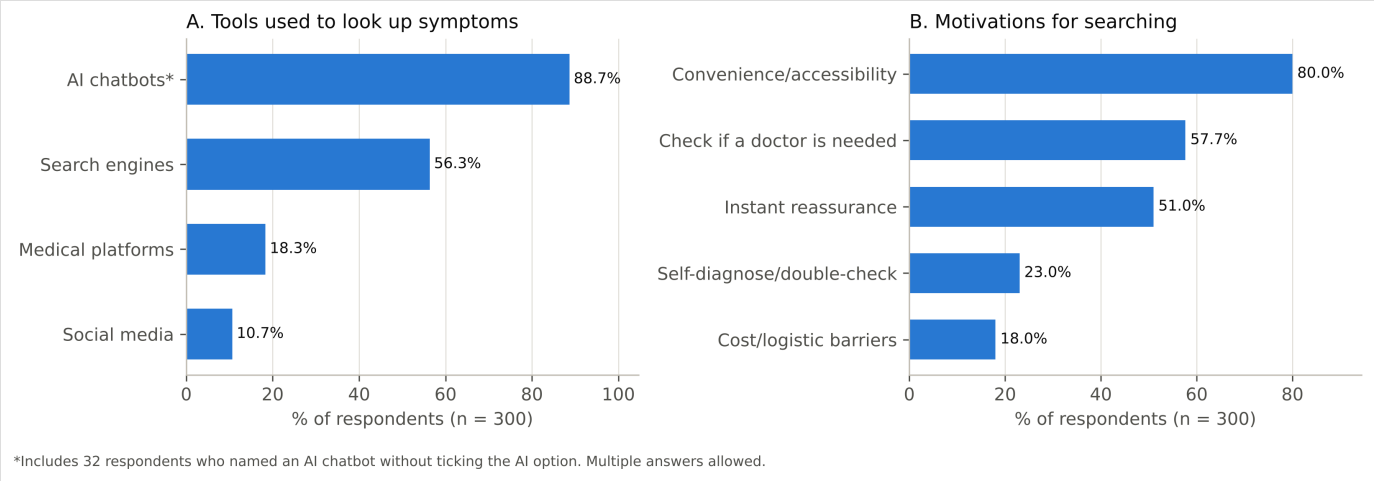


Figure 3: tools and motives. Read: bars = % of participants; multiple answers allowed, so they add to more than 100%.

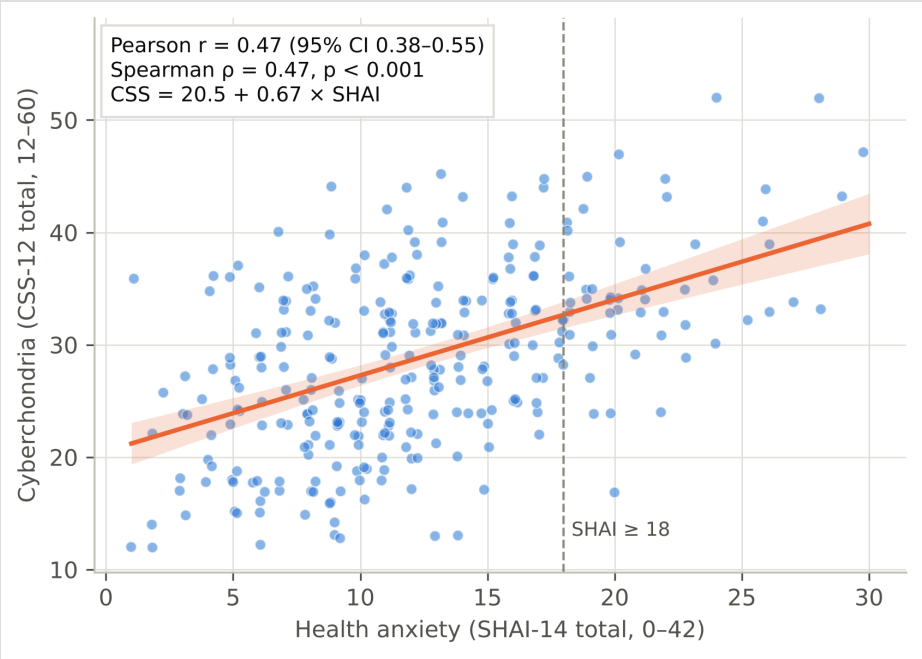


Figure 5: each dot is a participant; the line is the best fit, the band its 95% CI. The upward slope is r = 0.47.

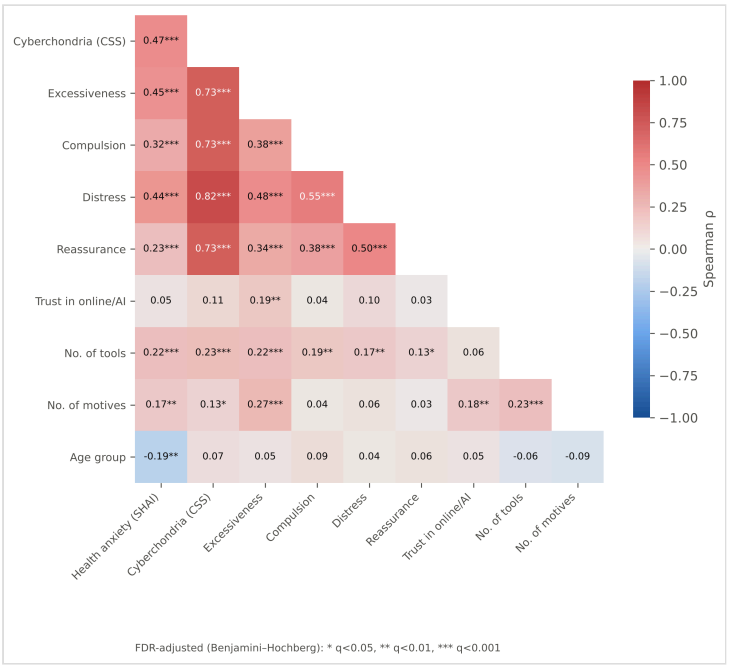


Figure 6: darker = stronger correlation; stars = significant after FDR.

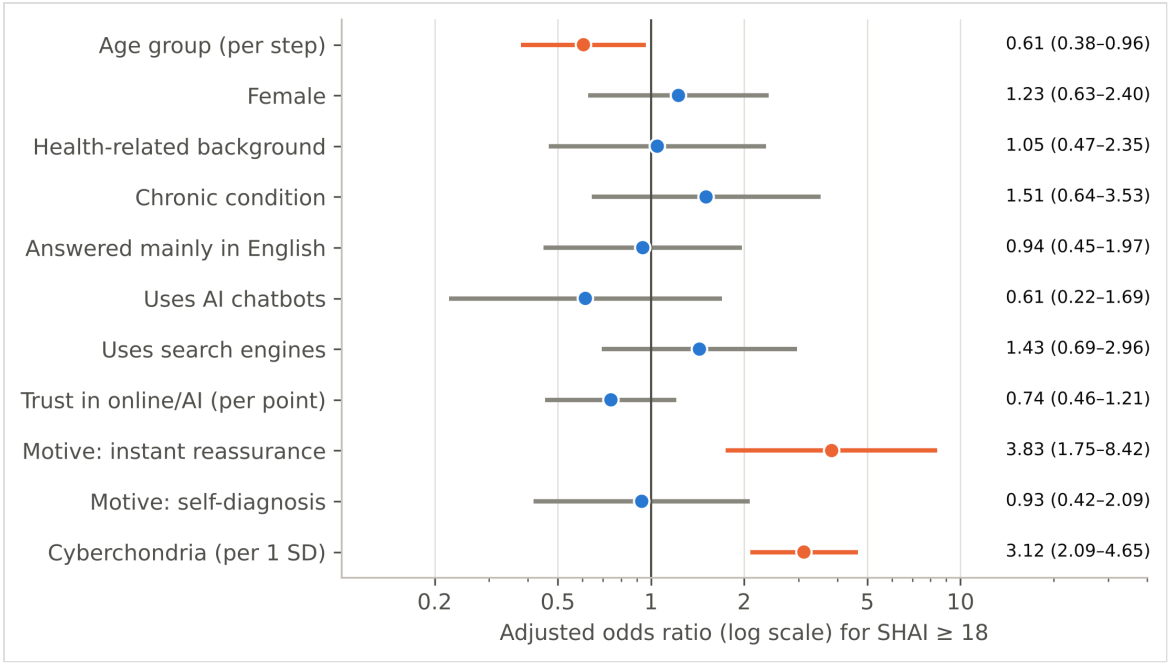


Figure 8: each dot is an AOR with its 95% CI; the vertical line at 1 = no effect; lines that cross 1 are not significant.

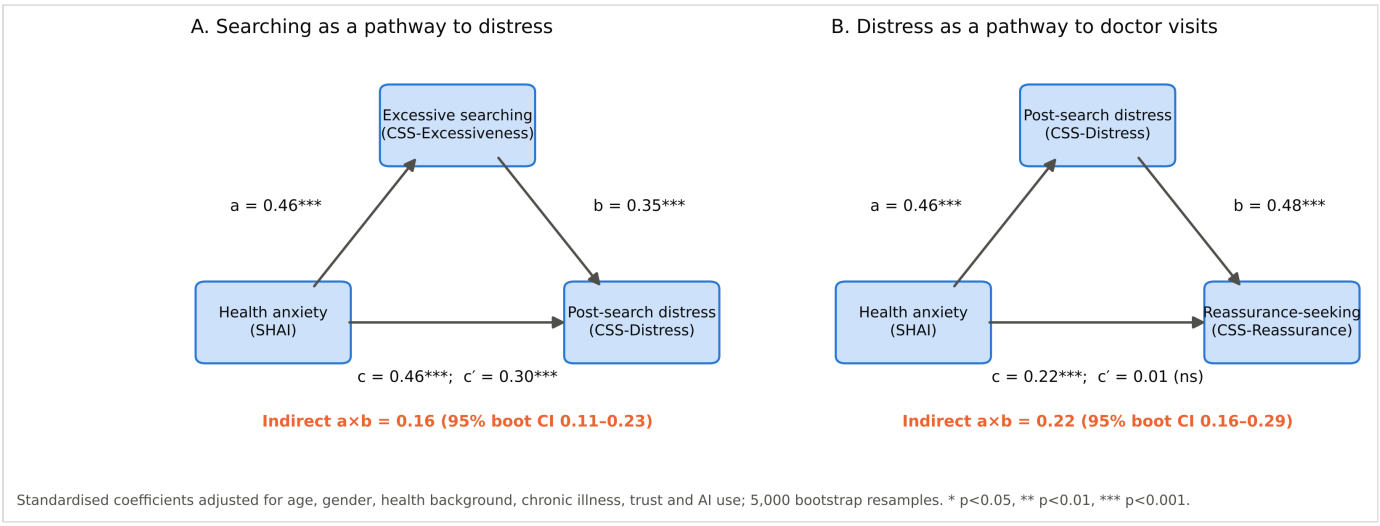


Figure 9: a = predictor → mediator, b = mediator → outcome, c′ = direct effect; the indirect effect is a × b.

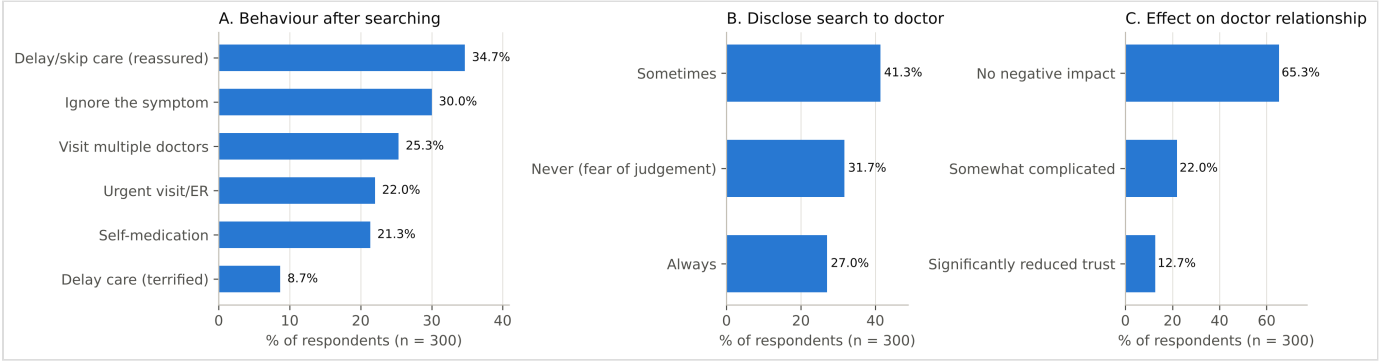


Figure 4: behaviour after searching, disclosure, and effect on the relationship.

Part 3 · Your key sources

Every reference cited in your chapter, with its number in the thesis. Know the bold ones well: examiners often ask “who found that?”.

No.	Reference (as listed in Chapter 6)
[14]	McElroy E, Kearney M, Touhey J, Evans J, Cooke Y, Shevlin M. The CSS-12: development and validation of a short-form version of the cyberchondria severity scale. Cyberpsychol Behav Soc Netw. 2019;22(5):330-5. doi:10.1089/cyber.2018…
[15]	Hallit S, Rogoza R, Abi Semaan C, Azzi V, Sawma T, Obeid S. Validation of the Arabic version of the cyberchondria severity scale 12 items (CSS-12-Ar) among a sample of Lebanese adults. BMC Psychiatry. 2023;23(1):618. doi:10.1186/s…
[18]	McMullan RD, Berle D, Arnáez S, Starcevic V. The relationships between health anxiety, online health information seeking, and cyberchondria: systematic review and meta-analysis. J Affect Disord. 2019;245:270-8. doi:10.1016/j.jad.2…
[19]	Schenkel SK, Jungmann SM, Gropalis M, Witthöft M. Conceptualizations of cyberchondria and relations to the anxiety spectrum: systematic review and meta-analysis. J Med Internet Res. 2021;23(11):e27835. doi:10.2196/27835
[37]	von Elm E, Altman DG, Egger M, Pocock SJ, Gøtzsche PC, Vandenbroucke JP. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. Lancet. 2007;370…
[38]	Eysenbach G. Improving the quality of Web surveys: the Checklist for Reporting Results of Internet E-Surveys (CHERRIES). J Med Internet Res. 2004;6(3):e34. doi:10.2196/jmir.6.3.e34
[39]	Serdar CC, Cihan M, Yücel D, Serdar MA. Sample size, power and effect size revisited: simplified and practical approaches in pre-clinical, clinical and laboratory studies. Biochem Med (Zagreb). 2021;31(1):010502. doi:10.11613/BM.2…
[40]	Salkovskis PM, Rimes KA, Warwick HMC, Clark DM. The Health Anxiety Inventory: development and validation of scales for the measurement of health anxiety and hypochondriasis. Psychol Med. 2002;32(5):843-53. doi:10.1017/S00332917020…
[41]	Arnáez S, García-Soriano G, López-Santiago J, Belloch A. The Spanish validation of the Short Health Anxiety Inventory: psychometric properties and clinical utility. Int J Clin Health Psychol. 2019;19(3):251-60. doi:10.1016/j.ijchp…
[42]	Österman S, Axelsson E, Lindefors N, Hedman-Lagerlöf E, Hedman-Lagerlöf M, Kern D, et al. The 14-item short health anxiety inventory (SHAI-14) used as a screening tool: appropriate interpretation and diagnostic accuracy of the Swe…
[43]	Poshtan MM, Aflakseir A, Witthöft M, Cherry MG, Ramzi M. Persian version of the Short Health Anxiety Inventory (SHAI) for patients with cancer: evaluation of psychometric properties, factor structure, and association with related …
[44]	Alshayea AK. Latent structure, measurement invariance, and reliability of an Arabic version of the Short Health Anxiety Inventory. J Exp Psychopathol. 2020;11(2). doi:10.1177/2043808720912629
[45]	Cruchinho P, López-Franco MD, Capelas ML, Almeida S, Bennett PM, Miranda da Silva M, et al. Translation, cross-cultural adaptation, and validation of measurement instruments: a practical guideline for novice researchers. J Multidi…
[46]	Alshaibani M. Validity and reliability of the short form of Cyberchondria Severity Scale (CSS-12) in Saudi Arabia. J Consum Health Internet. 2023;27(3):282-93. doi:10.1080/15398285.2023.2237347
[47]	Hayes AF, Coutts JJ. Use omega rather than Cronbach's alpha for estimating reliability. But… Commun Methods Meas. 2020;14(1):1-24. doi:10.1080/19312458.2020.1718629
[48]	McNeish D, Wolf MG. Dynamic fit index cutoffs for confirmatory factor analysis models. Psychol Methods. 2023;28(1):61-88. doi:10.1037/met0000425
[49]	Leitgöb H, Seddig D, Asparouhov T, Behr D, Davidov E, De Roover K, et al. Measurement invariance in the social sciences: historical development, methodological challenges, state of the art, and future perspectives. Soc Sci Res. 20…
[50]	Ćosić Pilepić A, Mohorić T, Takšić V. Multidimensional polytomous DIF detection methods: a Monte Carlo simulation study. Appl Psychol Meas. 2026;50(7):01466216261446320. doi:10.1177/01466216261446320
[51]	Steiger JH. Tests for comparing elements of a correlation matrix. Psychol Bull. 1980;87(2):245-51. doi:10.1037/0033-2909.87.2.245

[52]	Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. J R Stat Soc Series B Methodol. 1995;57(1):289-300. doi:10.1111/j.2517-6161.1995.tb02031.x
[53]	Hayes AF. Introduction to mediation, moderation, and conditional process analysis: a regression-based approach. 3rd ed. New York: Guilford Press; 2022.

Part 4 · Your slides, one by one

The script below is the speaker-notes text of your slides in the presentation. Practise until you can say it in your own words without reading.

Slide 13 · Results

Thank you. I will present the results in six steps: who took part, how they search, the levels of health anxiety and cyberchondria, the main association, the predictors and pathways, and finally behaviour and AI chatbots.

Slide 14 · Participants

Opening: Table 1 shows our 300 participants. Main finding: the sample is young, 62.3% aged 20 to 29, slightly more women at 56.3%, and 44.0% are medical students; 63.3% have a health-related background, and 21.7% reported a chronic condition. 70.7% answered mainly in Arabic. Closing: so this is a young, educated, digitally active sample, and that matters when we interpret percentages.

Slide 15 · How People Search

Opening: Figure 3 shows the tools people use and why. Main finding: AI chatbots are now the main gateway. 78.0% ticked AI chatbots, 88.7% if we include those who named one, compared with 56.3% for search engines. 41.0% use AI with no search engine at all, and ChatGPT alone is used by 59.3%. The top motives are convenience at 80.0%, checking whether a doctor is needed at 57.7%, and instant reassurance at 51.0%. Closing: in summary, for this sample, the first doctor is often an AI chatbot.

Slide 16 · Levels & Reliability

Opening: Table 3 shows the scores and reliability. Main finding: mean SHAI was 12.3 out of 42 and 19% scored 18 or more, our threshold for elevated health anxiety, with a 95% confidence interval of 15.0–23.8%. Mean CSS was 28.9 out of 60. Excessiveness was the highest facet and compulsion the lowest: people search a lot, but it rarely disrupts their lives. Both scales were reliable, alpha 0.81 and 0.86, and reliability was the same in Arabic and English responders. Formal invariance testing showed the CSS-12 works the same way in both languages, and no single item behaved differently. Closing: our tools measured well in this population.

Slide 17 · Health Anxiety ↔ Cyberchondria

Opening: this is our main result, Figure 5. Main finding: health anxiety and cyberchondria rise together, r = 0.47, 95% CI 0.38–0.55, so health anxiety explains about 22% of the variation in cyberchondria. After adjusting for every other variable, the partial correlation is still 0.48. The link is strongest for excessive searching, 0.46, and distress, 0.45, and weaker for compulsion and reassurance. Closing: in summary, hypotheses 1 and 2 are supported.

Slide 18 · What Predicts Cyberchondria?

Opening: Table 9 is the hierarchical regression for cyberchondria. Main finding: demographics alone explain only about 3% of cyberchondria. Adding digital-use variables brings it to 13%. Adding health anxiety adds another 18.4%, to about 32%. Health anxiety is the strongest predictor, standardised beta 0.48: each extra SHAI point adds about 0.68 CSS points. Closing: who you are matters little; how anxious you are matters most.

Slide 19 · Item by Item: What Drives Searching?

Opening: Figure 11 breaks the association down item by item, all 168 pairs. Main finding: 111 pairs remained significant after FDR correction. The strongest links involve fear of serious illness and interpreting body sensations; for example fear of illness with panicking when reading about a rare condition. Simply noticing aches and pains was almost unrelated. Closing: what drives searching is fear and interpretation, not just noticing symptoms, exactly what the cognitive model predicts.

Slide 20 · Who Has Elevated Health Anxiety?

Opening: Figure 8 shows who has elevated health anxiety, in a multivariable logistic model. Main finding: three factors are independent. Each standard deviation of cyberchondria triples the odds, adjusted odds ratio 3.12. Searching for instant reassurance almost quadruples them, 3.83. Older age is protective, 0.61 per age step. The model discriminates well: AUC 0.83, 0.79 after correcting for optimism. Closing: the person most at risk is young and searches to feel reassured.

Slide 21 · Pathways: Is There a Cycle?

Opening: Figure 9 shows the pathways. Main finding: in model A, part of the link from health anxiety to post-search distress runs through excessive searching, indirect effect 0.16, CI 0.11–0.23. In model B, post-search distress carries almost all of the link from health anxiety to reassurance-seeking visits: the direct effect drops to 0.01. But the reverse direction also fits, 0.15. Closing: our data are compatible with the cycle, but one survey at one time point cannot prove direction. Hypothesis 3 is supported statistically, not causally.

Slide 22 · Behaviour Before the Consultation

Opening: Figure 4 shows what people do after searching. Main finding: the most common response is to delay or skip care because they feel reassured, 34.7%. Others escalate: urgent visits 22.0%, multiple doctors 25.3%. Only 27.0% always tell their doctor about their searches, and 31.7% never do, for fear of being judged. Higher cyberchondria was linked with more disclosure, odds ratio 0.57, and this survived FDR correction. 34.7% say searching complicated or reduced trust in their doctor. Closing: searching changes the consultation before it begins.

Slide 23 · AI Chatbots: An Independent Effect?

Opening: finally, did AI chatbots themselves matter? Main finding: AI users scored 5.3 points higher on cyberchondria after adjustment, mainly excessiveness and reassurance. But 32 people named a chatbot without ticking the AI option. When we use the strict definition, only people who ticked it, the effect disappears: B = 1.5, p = 0.31. And only 34 people were true non-users. AI use also did not change the strength of the anxiety–cyberchondria link. Closing: no reliable independent effect of AI. Absence of evidence, not evidence of absence. I now hand over to my colleague for the discussion.

> **Delivery for your part:** ≈ 6 minutes; about 25–35 seconds per content slide and 10 seconds per divider. Point at the number you are saying. Pause after each key number. End with: “Those are our results. My colleague will now discuss what they mean.”

Part 5 · Questions and model answers

A. Questions on your part

Q: Why non-parametric tests?

Shapiro–Wilk showed the scores were not normally distributed; Mann–Whitney and Kruskal–Wallis do not assume normality. Welch tests confirmed the same conclusions.

Q: What is FDR and why did you use it?

With 162 comparisons, about 8 would be "significant" by chance at $p < 0.05$. Benjamini–Hochberg controls the proportion of false discoveries; results with $q \geq 0.05$ are labelled exploratory.

Q: Explain r = 0.47 to a non-statistician.

People with more health anxiety clearly tend to have more cyberchondria; anxiety accounts for about a fifth (22%) of the differences.

Q: Why both Pearson and Spearman?

Pearson for the linear correlation, Spearman as a check that does not depend on normality; both were 0.47.

Q: What does partial r = 0.48 tell us?

The link is not explained by age, gender, background, chronic disease, trust or AI use: it survives adjustment (H2).

Q: Why hierarchical regression?

To see what health anxiety adds after demographics and digital use: 18.4% extra variance, by far the biggest block.

Q: What do B and β mean in Table 9?

B = 0.68: each SHAI point adds about 0.7 CSS points. β = 0.48: the standardised effect, used to compare predictors.

Q: Why robust (HC3) standard errors?

The residuals were heteroskedastic (Breusch–Pagan $p = 0.004$); HC3 gives valid standard errors anyway.

Q: What is an adjusted odds ratio of 3.12?

For each standard deviation higher cyberchondria, the odds of elevated health anxiety are about three times higher, holding other factors constant.

Q: Why is the reassurance-motive AOR so wide?

Only 57 events for 11 predictors (≈ 5 per variable), so estimates are imprecise and probably optimistic; we label them exploratory.

Q: What is the AUC and why corrected?

How well the model separates elevated from non-elevated anxiety. Models look better on their own data, so 500 bootstrap refits corrected 0.83 to 0.79.

Q: Does the mediation prove a vicious cycle?

No. One time point, the reverse order fits too, and the facets come from one questionnaire. It is compatible with a cycle; only longitudinal data can show direction.

Q: Why does the AI effect disappear?

It depends on 32 people who named a chatbot without ticking it; they had the highest scores. Under the strict definition $p = 0.31$. Only 34 non-users made a small comparison group.

Q: Chronic illness was not significant in the logistic model. Why?

Dichotomising SHAI at 18 loses information and power. On the continuous score chronic illness is significant: $p = 0.004$ unadjusted and $p = 0.029$ adjusted (B +1.7).

Q: What do CFI, RMSEA and SRMR mean?

Fit indices for CFA. CSS-12 four factors: CFI 0.95, RMSEA 0.070, SRMR 0.053: acceptable. SHAI one-factor fitted poorly.

Q: Why does the SHAI fit poorly?

It is not strictly unidimensional; recent validations find two factors (illness likelihood and consequences). Reliability was still good (α 0.81).

Q: What is measurement invariance (Table 5)?

Testing in steps whether the scale works the same in Arabic and English: same structure, then same loadings, then same intercepts. CSS-12 passed all three; SHAI-14 passed the first two.

Q: Why is high cyberchondria defined as > 33?

There is no validated clinical cut-off for the CSS-12, so we used the upper tertile of our sample, as is common.

Q: Why is the prevalence so dependent on the cut-off (Table 4)?

Because most scores cluster just below the thresholds; that is why we report several cut-offs and say "elevated", not "disorder".

Q: What does Table 13 show?

That the main correlation stays 0.46–0.48 whatever we exclude, and the regression is stable. Only the AI effect depended on a decision.

B. Questions anyone can be asked (know the answer even if it is not your part)

Q: Summarise your study in one minute.

Use the two-minute script in Part 1, shortened: design, sample, two scales, r = 0.47, AI result, behaviour, clinical message.

Q: What is your main finding?

Health anxiety and cyberchondria are moderately and robustly associated (r = 0.47), independent of other factors.

Q: What did you find about AI chatbots?

They are the main source (78.0% ticked), but there was no reliable independent effect on cyberchondria beyond health anxiety.

Q: What percentage had elevated health anxiety?

19% at SHAI ≥ 18, a screening threshold, not a diagnosis.

Q: What is your biggest limitation?

The cross-sectional design: no direction of cause.

Q: What is the clinical message?

Ask patients, without judgement, what they searched or asked an AI.

Q: Why did you split the presentation this way?

By chapter, but each of us studied the whole thesis.

Q: Were your hypotheses written before you collected the data?

Answer truthfully. A safe true answer: they follow from our objectives and previous research, and the questionnaire was designed to measure these things; we tested all of them and report every result, including those not supported.

Watch out: Never claim they were pre-registered.

Q: Who did the statistical analysis?

Answer honestly according to what you did. If AI tools helped with code or drafting, say so in line with your college policy and the declaration in the thesis.

C. Difficult or hostile questions

Q: "Your p-values are tiny because your sample is large."

The effect sizes are meaningful too: r = 0.47 is moderate and ΔR^2 18.4% is a medium effect ($f^2 \approx 0.27$).

Q: "You ran so many tests; something will be significant."

Exactly why we used FDR correction and separated robust from exploratory findings.

Q: "The mediators are subscales of the same scale. Isn't that circular?"

Partly, yes: shared method variance inflates the paths. That is why we call it an exploratory decomposition, not a mechanism.

D. Limitations and what was done about each (all students)

Limitation	What we did / how to answer
Cross-sectional design	Reverse-direction models; wording "compatible with"; longitudinal studies recommended
Convenience/snowball sample	Percentages not generalised; associations consistent across subgroups
Arabic wording (no back-translation)	Equal reliability and means; CSS-12 fully invariant, SHAI-14 metric; no item DIF
Single-item behaviour measures	Reported descriptively; associations labelled exploratory
Many tests	FDR correction; robust vs exploratory separated
Small subgroups (34 non-users)	Acknowledged; AI effect called "not robust"
Self-report and careless responding	Anonymous; straight-liners and annotated responses excluded in sensitivity analyses (no change)

> **How to answer any question:** pause → repeat the question briefly → answer in two or three sentences → stop. If you do not know: "We did not examine that directly; based on our data we would expect…, and it is a good question for future research." Never invent a number. If a teammate struggles: "If I may add…".

Part 6 · Editing your chapter

> **The Word file is the master copy.** Edit only the .docx, in one shared OneDrive / Word Online file with **Review → Track Changes** on. Keep an untouched copy (Thesis_ORIGINAL.docx). Avoid Google Docs: it breaks the automatic numbering.

For your chapter

- Results is mostly numbers: **do not change any number**, p, q, CI, test name or table reference.
- You may reword the connecting sentences in your own words; keep one main message per paragraph.
- Never type table or figure numbers by hand in captions; in text, keep "Table n / Figure n" exactly.
- If a number or sentence looks wrong, add a comment and ask; do not "fix" statistics yourself.
- Read 2.6 Statistical analysis until you can explain each method in one sentence; you will be asked.

Never touch

- Numbers, %, p, q, CI, r, OR, β, n
- Reference numbers like [18] (move them with their sentence)
- The reference list
- Table/figure captions, table of contents and lists (automatic fields)
- Heading numbers and styles
- Times New Roman 14 and page numbering

The rewrite method (paragraph by paragraph)

1. Read until you understand it; look up anything unclear.
2. Close it and explain it out loud.
3. Write your version from that explanation.
4. Compare: same facts, numbers and [n]; nothing claimed more strongly.
5. Read aloud; change any phrase you would never say.

After editing

1. Ctrl + A, then F9, to update the table of contents, the lists and the captions.
2. Search for "[" to find placeholders left.
3. Swap chapters with a teammate for checking.
4. Export to PDF and look at every page.

> **About AI use, honestly.** Check your college's AI policy with your supervisor and declare AI assistance if required. A sentence you can adapt: *"The statistical analysis code and an initial draft of this report were prepared with the assistance of an AI tool (Claude, Anthropic). All analyses were checked, and the text was reviewed and rewritten by the authors, who take full responsibility for its content."* Do not use paraphrasing tools or tricks to fool detectors: they make the text worse and can count as misconduct. Real ownership (your words, your understanding, your edit history) is the best protection.

Part 7 · Your 14-day plan

Day	Task
1	Read the Abstract, 1.8 and Chapter 5. Learn the two-minute script (Part 1).
2	Read the whole thesis once, marking what you cannot explain.
3	Core facts and terms (Part 1). Self-test on core facts.
4	Your chapter, sections in Part 2 (first half).
5	Your chapter, second half; your key sources (Part 3).
6	Start rewriting your chapter (Part 6).
7	Finish rewriting. Self-test (Part 8).
8	Swap chapters with a teammate; check accuracy and voice.
9	Your slides: read the scripts (Part 4), then present them alone 3 times.
10	Questions A–D out loud (Part 5). Record yourself once.
11	Team: full rehearsal with timing and handovers.
12	Team: mock committee; each student asks the others their Part 5 questions.
13	Fix weak answers; second full rehearsal; final Word checks.
14	Day before: checklist (last page); one calm run-through; sleep.

Part 8 · Self-test

Cover the answers, answer out loud, then check. Repeat until all are correct.

#	Question	Answer
1	Main correlation with 95% CI?	r = 0.47 (0.38–0.55)
2	Partial correlation?	0.48
3	Which two facets correlate most with health anxiety?	Excessiveness (0.46) and distress (0.45)
4	R^2 of the three regression blocks?	0.030 → 0.133 → 0.317
5	β and B for health anxiety in the final model?	β 0.48, B 0.68
6	Three independent predictors of elevated health anxiety and their AORs?	Cyberchondria 3.12, reassurance motive 3.83, age 0.61
7	AUC apparent and corrected?	0.83 → 0.79
8	Indirect effect via excessive searching?	0.16 (CI 0.11–0.23)
9	Direct effect of HA on reassurance seeking after distress?	0.01 (not significant)
10	The only behavioural association that survived FDR?	Higher cyberchondria → less never-disclosure (AOR 0.57, q 0.012)
11	AI effect inclusive vs strict?	+5.3 vs +1.5 (p 0.31)
12	Prevalence at cut-offs 15/18/22/29?	32.7% / 19.0% / 7.3% / 0.7%
13	Which scale is fully invariant across languages?	CSS-12 (scalar); SHAI-14 metric only
14	How many of 168 item pairs survived FDR?	111
15	Reliability α of both scales?	0.81 and 0.86
16	Main correlation (everyone)?	r = 0.47
17	Elevated health anxiety (everyone)?	19%

18	AI chatbots ticked (everyone)?	78.0%
19	Never tell the doctor (everyone)?	31.7%
20	Sample size (everyone)?	300

Part 9 · Day-before checklist

✓	Item
☐	All placeholders filled (names, supervisor, college, logo, degree, year, ethics, Google Forms question, Acknowledgment)
☐	Fields updated (Ctrl + A, F9); PDF checked page by page
☐	AI declaration added if required
☐	Presentation: names on slide 1, ethics number on slide 10; USB + PDF + email copies; clicker tested
☐	I can give the two-minute summary and my slides (13–23) without notes
☐	I have answered every question in Part 5 out loud
☐	I scored 100% on the self-test
☐	Printed thesis copy if required; arrive early; agree who stands where

Generated from the study outputs and the presentation, so every number and slide script here matches the thesis and the deck.